

Xiaoyang Chen

Ph.D. Student in Mathematics · Tsinghua university

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Education

Incoming Ph.D. in Mathematics (expected 2031) <i>Qiuzhen college, Tsinghua university, Beijing, China</i> Expected advisor: Prof. Penghui Li. Intended research area: Geometric representation theory.	Starting September 2026
B.Sc. in Mathematics <i>Qiuzhen college, Tsinghua university, Beijing, China</i>	2023–2026

Research Interests

Geometric representation theory; geometric Langlands; graded sheaves; derived algebraic geometry; higher category theory.

Research Experience

Undergraduate Thesis — “Langlands duality of affine Hecke categories” <i>Qiuzhen college, Tsinghua university</i> Advisor: Prof. Penghui Li. The paper gives a graded lift of the Bezrukavnikov-Arkhipov equivalence.	2025–2026
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Seminar Talks and Presentations

The geometric Satake equivalence co-speakers: Mingyu Bai, Kun Zheng	December 2025
Equivariant K-theory and index theory	June 2025
The moduli stack of formal groups and chromatic homotopy theory	May 2025
Galois extensions of structured ring spectra and Goerss–Hopkins theory	Month 2025
The notes of these talks are on my website	

Conferences and Workshops

International Workshop on Algebraic Topology 2026 (IWoaT 2026) <i>Peking university, Beijing</i> participant	August 2026
International Congress of Mathematicians 2026 (ICM 2026) <i>IMU, Philadelphia</i> participant	July 2026
International Congress of Chinese Mathematicians 2025 (ICCM 2025) <i>SIMIS, Shanghai</i> participant	January, 2026
Beijing–Shanghai Summer School in Mathematics 2025 <i>SIMIS, Shanghai</i> participant, the topic of this conference is the geometric Langlands program	July 2025
International Congress of Basic Science <i>BIMSA, Beijing</i> participant	July 2023, July 2024, July 2025, August 2026

Fellowships, Scholarships, and Awards

Undergraduate thesis award	June 2026
<i>Qiuzhen College, Tsinghua university</i>	
Fellowship 2024	2024
<i>Qiuzhen College, Tsinghua university</i>	